

EMPIRICA

Epistemic Self-Assessment for AI Agents

What is Empirica?

Empirica is an epistemic self-assessment system that helps AI agents (and humans) track what they know and learn. It quantifies knowledge across 13 vectors organized into 3 tiers.

Core Concept

Before work: **What do I know?** (Preflight)

After work: **What did I learn?** (Postflight)

The difference = Quantified learning

The 13 Epistemic Vectors

Tier 0: Foundation

engagement	How focused/committed to the task
know	Factual knowledge about the domain
do	Procedural knowledge (how to)
context	Understanding of surrounding situation

Tier 1: Comprehension

clarity	How clear the understanding is
coherence	How well pieces fit together
signal	Signal-to-noise ratio
density	Information richness

Tier 2: Execution

state	Current system state awareness
change	Understanding of what changed
completion	Task completion level
impact	Effect of actions taken

Meta

uncertainty	Explicit tracking of unknowns
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The CASCADE Workflow

PREFLIGHT → WORK → CHECK → POSTFLIGHT

1. Preflight (Before Work)

Assess what you know before starting:

```
1 echo '{
2   "session_id": "abc-123",
3   "vectors": {
4     "engagement": 0.8,
5     "foundation": {"know": 0.6, "do": 0.7, "context": 0.5},
6     "comprehension": {"clarity": 0.7, "coherence": 0.8},
7     "uncertainty": 0.4
8   },
9   "reasoning": "Starting with moderate OAuth2 knowledge..."
10 }' | empirica preflight-submit -
```

Shell

2. Work (Do the Task)

Log findings and unknowns as you work:

```
1 empirica finding-log --finding "Discovered token refresh" --impact 0.7
2 empirica unknown-log --unknown "Need to investigate scopes"
```

Shell

3. Check (Decision Points)

At critical junctures, run a check gate:

```
1 empirica check-submit -
```

Shell

Returns: PROCEED | HALT | BRANCH | REVISE

4. Postflight (After Work)

Measure what you actually learned:

```
1 echo '{
2   "session_id": "abc-123",
3   "vectors": {
4     "foundation": {"know": 0.85, "do": 0.9},
5     "uncertainty": 0.15
6   },
7   "reasoning": "Successfully implemented OAuth2"
8 }' | empirica postflight-submit -
```

Shell

Result: know: +0.25, uncertainty: -0.25 (quantified learning!)

WAPT Integration

Session Management

 Shell

Logging Commands

 Shell

Dashboard

 Shell

Gamification

Empirica includes RPG elements to make epistemic tracking engaging:

Character Sheet

1	
2	EPISTEMIC ADVENTURER - Level 5
3	
4	STR: 14 INT: 16 WIS: 12
5	DEX: 10 CON: 13 CHA: 11
6	
7	XP: 2340/3000 Karma: +42
8	

Moon Phase Health Indicators

-  Critical (coverage < 25%)
-  Low (25-50%)
-  Moderate (50-75%)
-  Good (75-90%)
-  Excellent (90%+)

Why This Matters

Traditional AI development doesn't track **what the AI knows**. Empirica changes that:

- **Accountability:** Know what was known when decisions were made
- **Learning:** Quantify actual knowledge gain
- **Safety:** Explicit uncertainty tracking prevents overconfidence
- **Science:** Rigorous data for AI cognition research